

Experiment No : 4

Title :Implement Database Management using arrays of structures with operations Create, Display, Modify,Search and Sort.

Name of Student:Shivam Sevewar

Class: E&TC

Div: SY1

PRN No:B25ET1042

Batch:B

Program:

```
#include <stdio.h>
```

```
#include <string.h>
```

```
struct student
```

```
{  
    char name[20];  
    int roll_num;  
} s[100];
```

```
void create(int n)
```

```
{  
    int i;  
  
    for(i = 0; i < n; i++)  
    {  
        printf("\nEnter name: ");  
        scanf("%s", s[i].name);  
  
        printf("Enter roll no.: ");  
        scanf("%d", &s[i].roll_num);  
    }  
}
```

```
void display(int n)
```

```
{  
    int i;  
  
    for(i = 0; i < n; i++)
```

```
{
    printf("\nName: %s", s[i].name);
    printf("\nRoll No.: %d\n", s[i].roll_num);
}
}
```

void modify(int n)

```
{
    int roll, i;

    printf("\nEnter roll no. to modify: ");
    scanf("%d", &roll);

    for(i = 0; i < n; i++)
    {
        if(s[i].roll_num == roll)
        {
            printf("Enter new name: ");
            scanf("%s", s[i].name);

            printf("Enter new roll no.: ");
            scanf("%d", &s[i].roll_num);

            printf("Record modified successfully.\n");
            return;
        }
    }

    printf("Record not found.\n");
}
```

void search(int n)

```
{
    int roll, i;
```

```
printf("\nEnter roll no. to search: ");
```

```
scanf("%d", &roll);
```

```
for(i = 0; i < n; i++)
```

```
{
```

```
    if(s[i].roll_num == roll)
```

```
    {
```

```
        printf("\nRecord found!");
```

```
        printf("\nName: %s", s[i].name);
```

```
        printf("\nRoll No.: %d\n", s[i].roll_num);
```

```
        return;
```

```
    }
```

```
}
```

```
printf("\nRecord not found.\n");
```

```
}
```

```
void sort(int n)
```

```
{
```

```
    int i, j;
```

```
    struct student temp;
```

```
    for(i = 0; i < n - 1; i++)
```

```
    {
```

```
        for(j = i + 1; j < n; j++)
```

```
        {
```

```
            if(s[i].roll_num > s[j].roll_num)
```

```
            {
```

```
                temp = s[i];
```

```
                s[i] = s[j];
```

```
                s[j] = temp;
```

```
            }
```

```
        }
```

```
    }
```

```
    printf("\nRecords sorted successfully.\n");  
}
```

```
int main()
```

```
{  
    int n, choice;
```

```
  
    printf("Enter number of students: ");  
    scanf("%d", &n);
```

```
  
    do
```

```
{  
    printf("\n\n1. Create");  
    printf("\n2. Display");  
    printf("\n3. Modify");  
    printf("\n4. Search");  
    printf("\n5. Sort");  
    printf("\n6. Exit");
```

```
  
    printf("\nEnter your choice: ");  
    scanf("%d", &choice);
```

```
  
    switch(choice)
```

```
{  
    case 1:  
        create(n);  
        break;
```

```
  
    case 2:  
        display(n);  
        break;
```

```
  
    case 3:  
        modify(n);  
        break;
```

```
    case 4:
        search(n);
        break;

    case 5:
        sort(n);
        break;

    case 6:
        printf("\nExiting...");
        break;

    default:
        printf("\nInvalid choice!");
}

} while(choice != 6);

return 0;
}
```

Output:

Output

Enter number of students: 3

1. Create
2. Display
3. Modify
4. Search
5. Sort
6. Exit

Enter your choice: 1

Enter name: Anurag

Enter roll no.: 18

Enter name: Shivam

Enter roll no.: 45

Enter name: Sanchit

Enter roll no.: 7